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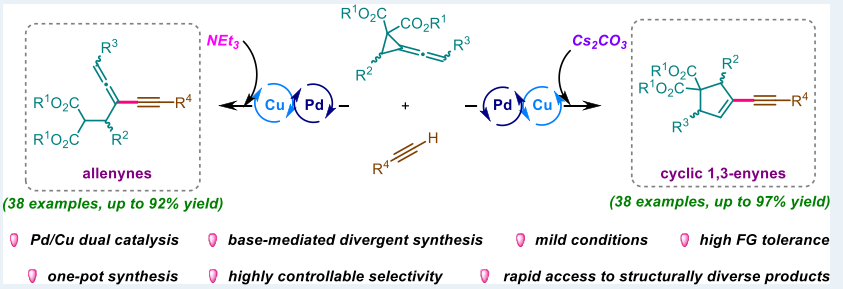
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"Cesium Effect"-Mediated/Controlled Divergent Synthesis via Synergistic Pd/Cu-Catalyzed Cascade Reactions

Xian-Hua Zhong, Yong-Jie Long, Yin Wei,* and Min Shi*

Cite This: ACS Catal. 2025, 15, 8390−8402 Read Online

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ABSTRACT: Herein, we describe a method in the one-pot synthesis of cyclic 1,3-enynes and allenynes from alkynyl-allenyl cross-coupling and cascade reactions of vinylidenecyclopropane-diesters (VDCP-diesters) with a large number of unactivated terminal alkynes for the formation of a C(sp^2)−C(sp) bond under palladium/copper dual catalysis and assisted by base Cs_2CO_3 and NEt_3 with distinctly different Brønsted basicity and Lewis basicity. This developed divergent transformation proceeds through a key rapid transmetalation and the subsequent intramolecular cyclization process regulated by cesium effect. Our investigations found that these reactions can occur under mild conditions and have a broad substrate scope with a wide range of functional groups via a zwitterionic palladium species, affording the corresponding desired products in moderate to good yields. Further synthetic transformations show the application value of this protocol. Plausible reaction mechanisms have also been proposed on the basis of control, deuterium labeling experiments, and density functional theory calculations.

KEYWORDS: *C(sp^2)−C(sp) cross-coupling, palladium/copper dual catalysis, cesium effect, divergent synthesis, zwitterionic palladium species*

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INTRODUCTION

As the precursors and high value-added derivatives of allenes, cyclic 1,3-enynes and allenynes serve as key components in a variety of functional materials and act as unique reactive intermediates in organic synthesis.[1−12] A large number of literature have reported that these compounds not only possess important physiological and pharmacological activities but also have plentiful reactivities for the synthesis of natural products and drug-like molecules (Scheme 1A).[1,13−20] However, the lack of efficient and versatile synthetic methods has significantly hindered their broader application in synthetic chemistry and biological studies. Consequently, considerable efforts have been devoted to exploring novel methodologies for their efficient syntheses over the past few decades. The current synthesis of cyclic 1,3-enynes and allenynes typically involves palladium-catalyzed cross-coupling reactions to construct the new C(sp^2)-C(sp) bond, where terminal alkynes or alkynyl organometallic reagents are used to react with alkenyl or allenic halides.[21−27] Nevertheless, this process typically necessitates the introduction of appropriate leaving groups onto alkenyl, allenyl, or propargyl moieties and does not ensure compatibility with other functional groups (Scheme 1B). Therefore, upon synthesizing cyclic 1,3-enynes and allenynes with unique structures, it remains essential to identify appropriate alternative methods.

Nonetheless, several challenges remain to be resolved in the formulation of a comprehensive strategy for their synthetic methods. For example, these methods should feature rapid synthesis, a broad substrate scope, mild reaction conditions, high reaction yields, and excellent atom economy. In this research aspect, Ma,[28−33] Liu,[34] Han and Shen,[35] Tunge,[36] and Wu, Bao, Guo, and Qiu[37] have accomplished a variety of excellent work on the development of elegant strategies for the synthesis of allenynes and the related substances, respectively. However, the synthetic approaches reported to date have yet to achieve the synthesis of 1,1-disubstituted allenynes that can

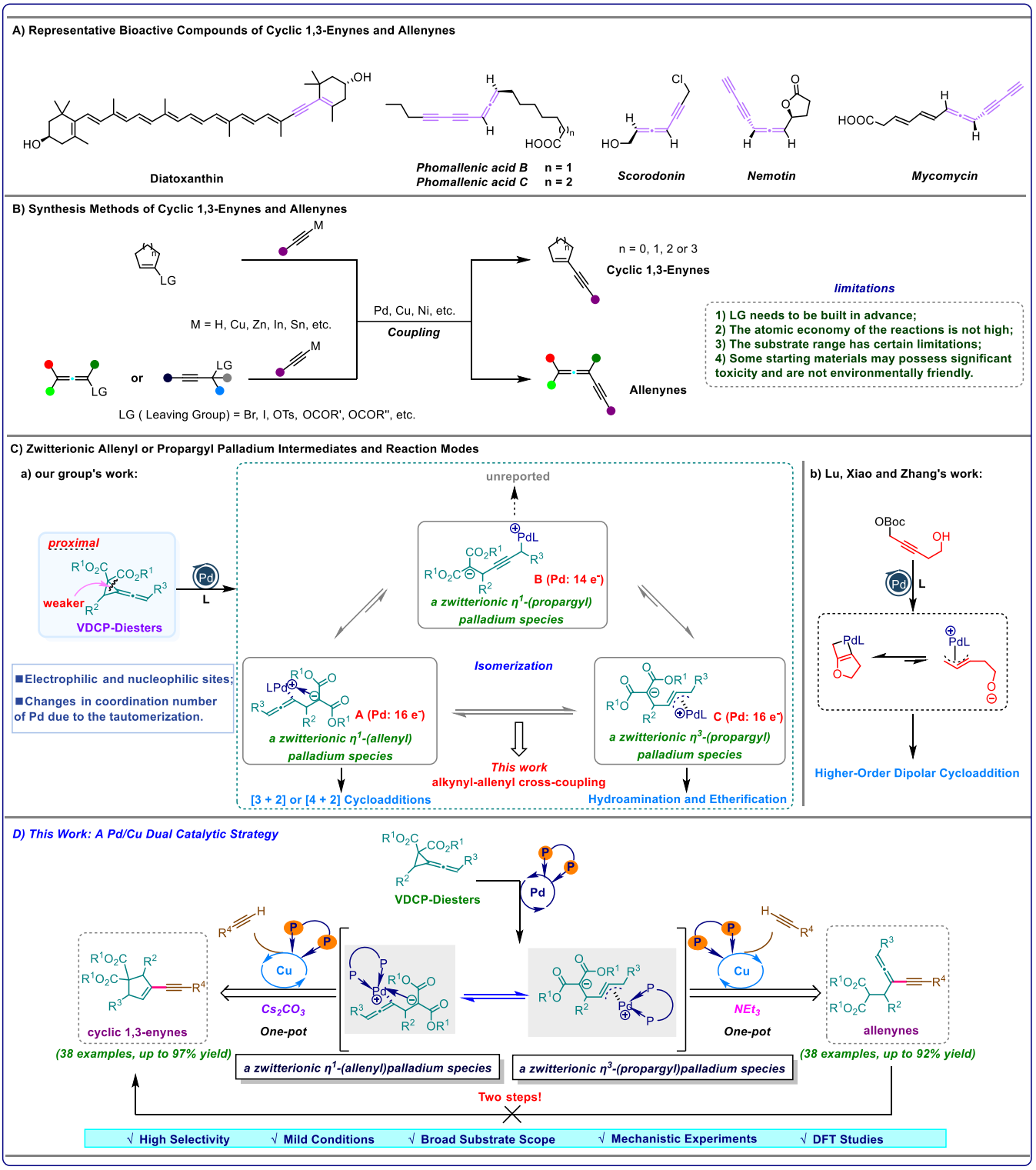
Received: March 17, 2025
Revised: April 23, 2025
Accepted: April 24, 2025
Published: May 5, 2025



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https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Scheme 1. Research Background, Previous Works, and This Work. (A) Representative Bioactive Compounds of Cyclic 1,3-Enynes and Allenynes. (B) Synthetic Methods of Cyclic 1,3-Enynes and Allenynes. (C) Zwitterionic Allenyl or Propargyl Palladium Intermediates and Reaction Modes. (D) A Pd/Cu Dual Catalytic Strategy of VDCP-Diesters with Terminal Alkynes

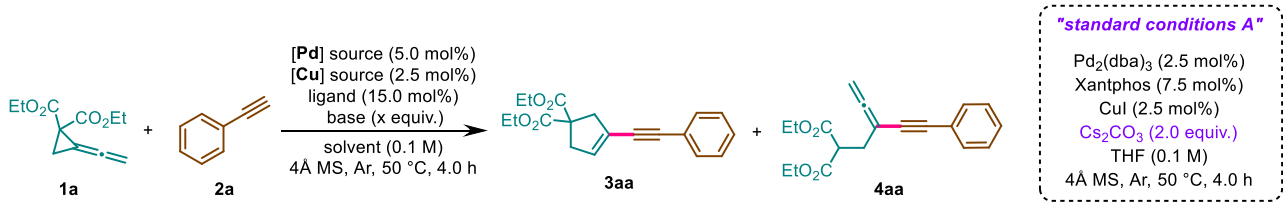


possess more reactive activity, indicating that the exploration of alternative methodologies to address this gap is still necessary.

On the other hand, compared to reactions involving π-allyl palladium species, palladium-catalyzed couplings of allenyl or propargyl electrophiles have received considerably less attention.[38,39] Moreover, as the crucial and significant intermediate, zwitterionic allenyl palladium dipoles have received limited attention in the literature as well.[40] A potential reason for the lesser development of these couplings is the increased complexity in achieving regioselective and chemoselective substitution.[41,42] Specifically, achieving chemoselectivity and regioselectivity is more challenging in

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Table 1. Optimization of Reaction Conditions



entry[a]	deviation from standard conditions A	yield of **3aa** (%)[b]	yield of **4aa** (%)[b]
1[c]	10.0 mol% of $Sc(OTf)_3$ was used, 1.0 equiv. of Cs_2CO_3	11	20
2[d]	without Cs_2CO_3	trace	trace
3[d]	1.0 equiv. of Cs_2CO_3	26	17
4[d]	2.0 equiv. of Cs_2CO_3	70	13
5[d]	3.0 equiv. of Cs_2CO_3	72	< 5
6[e]	"$Pd(OAc)_2$ and PPh_3" instead of "$Pd_2(dba)_3$ and Xantphos"	52	12
7[f]	"$Pd(OAc)_2$" instead of "$Pd_2(dba)_3$ and Xantphos"	trace	trace
8	**none**	***85 (83)***	***trace***
9	**standard conditions B: NEt_3 instead of Cs_2CO_3**	***trace***	***90 (82)***
10[g]	Ph—≡—Cu instead of CuI under standard conditions A	82	nd
11[g]	Ph—≡—Cu instead of CuI under standard conditions B	nd	72
12[h]	$AgNTf_2$ instead of CuI under standard conditions A	34	nd
13[i]	PPh_3AuNTf_2 instead of CuI under standard conditions A	44	8
14	without CuI	nd	nd
15	without $Pd_2(dba)_3$	nd	nd
16	without Xantphos	trace	trace

[a]Unless otherwise indicated, all reactions were carried out with **1a** (0.20 mmol), **2a** (0.30 mmol), [**Pd**] source (0.01 mmol), ligand (0.03 mmol), [**Cu**] source (0.005 mmol), base (*x* mmol), and 4 Å MS (100 mg) in dry THF (2.0 mL) at 50 °C for 4.0 h under an Ar atmosphere. [b]Yields were determined by ^1H NMR spectroscopic data using CH_2Br_2 as an internal standard. nd = not detected. Isolated yields are in parentheses. [c]5.0 mol % of $Pd(PPh_3)_4$ was used. 10.0 mol % of $Sc(OTf)_3$ was used as an additive. [d]5.0 mol % of $Pd(PPh_3)_4$ was used. [e]5.0 mol % of $Pd(OAc)_2$ and 15.0 mol % of PPh_3 were used. [f]5.0 mol % of $Pd(OAc)_2$ was used. [g]2.5 mol % of copper(I) phenylacetylide was used. [h]2.5 mol % of $AgNTf_2$ was used instead of [**Cu**] source. [i]2.5 mol % PPh_3AuNTf_2 was used instead of [**Cu**] source.

palladium-catalyzed allenyl or propargyl substitution reactions.[43−50]

It can be noted that this zwitterionic propargyl palladium intermediate usually exists in three forms, including η^1-(allenyl)palladium species, η^1-(propargyl)palladium, and η^3-(propargyl)palladium.[51,52] In the studies reported to date, the reaction methods employed to capture these active intermediates have predominantly been dipolar cycloaddition and nucleophilic addition. Our group has published several studies based on vinylidenecyclopropane-diesters (VDCP-diesters) as a zwitterionic allenyl palladium dipolar precursor in recent years (Scheme 1C(a)).[53−58] As an interesting special class of allenic derivatives, VDCP-diesters are the molecules with high tension but are relatively stable at room temperature (Scheme 1C(a)).[59] Additionally, Lu, Xiao, Zhang, and their co-workers have developed a series of propargyl palladium dipolar intermediates and reported a higher-order dipolar cycloaddition strategy in 2024 (Scheme 1C(b)).[60]

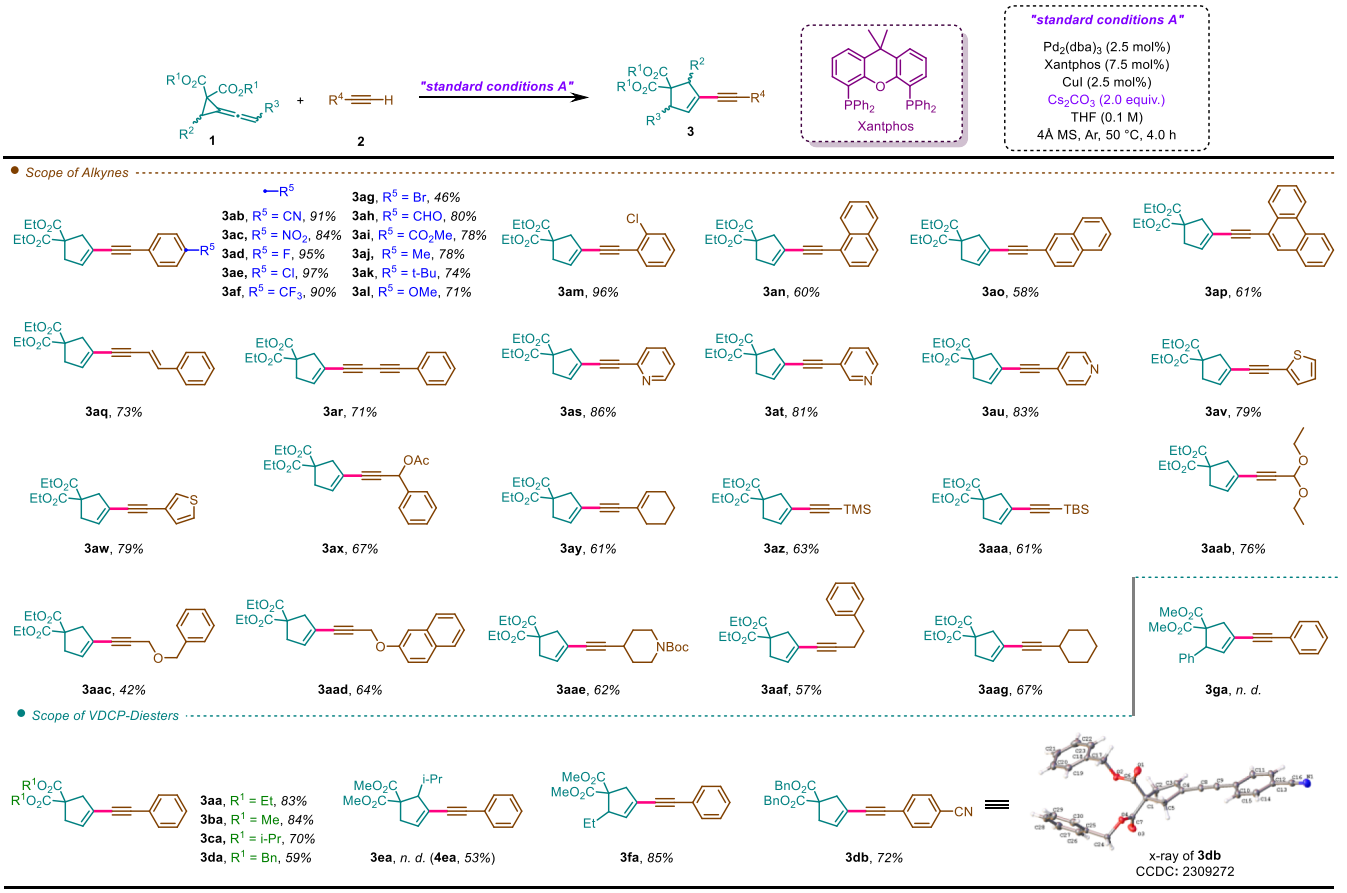
On the basis of these previous works, we attempted to carry out further investigation on the development of a new reaction mode of zwitterionic allenyl palladium intermediates different from existing reaction modes. In this paper, we wish to report a Pd/Cu bimetallic synergistic catalysis and cesium effect-mediated/controlled cascade reactions based on VDCP-diesters, which induced a divergent synthetic method of cyclic 1,3-enynes and allenynes that are obtained under the regulation of different bases (Scheme 1D). In contrast to the conventional reaction method, this reaction achieves in situ capture of the zwitterionic allenyl palladium species via a rapid transmetalation process, thereby obtaining the crucial intermediates. This protocol not only realizes a newly controllable transformation of strained small ring connected allenes to the other key intermediates[61−63] but also provides a reference idea for the efficient synthesis of conjugated moiety.

METHODS

General Procedure for Synergistic Pd/Cu-Catalyzed Coupling Reactions of Vinylidenecyclopropane Diesters with Terminal Alkynes. A dried 10 mL sealed tube was charged with $Pd_2(dba)_3$ (0.005 mmol, 2.5 mol %), Xantphos (0.015 mmol, 7.5 mol %), and 4 Å MS (100 mg) in dry tetrahydrofuran (THF) (2.0 mL) under the protection of argon, and then the mixture was stirred at ambient temperature for 30 min. After that, the reaction tube was evacuated and backfilled with argon 3 times, followed by the addition of VDCP-diesters **1** (0.2 mmol, 1.0 equiv), Cs_2CO_3 or dry NEt_3 (0.4 mmol, 2.0 equiv), CuI (0.005 mmol, 2.5 mol %), and terminal alkynes **2** (0.3 mmol, 1.5 equiv). Then, the reaction mixture was stirred at 50 °C. After 4.0 h, the reaction mixture was filtered through a pad of Celite and silica (eluent: hexane/dichloromethane/ethyl acetate = 5:1:1, v/v/v), then the filtrate was concentrated under reduced pressure, and the residue was

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Scheme 2. Substrate Scope of Cross-Coupling Reactions of Vinylidenecyclopropane-Diesters 1 with Terminal Alkynes 2 under the Standard Conditions A



[a]All reactions were carried out with **1** (0.2 mmol), **2** (0.3 mmol), $Pd_2(dba)_3$ (0.005 mmol), Xantphos (0.015 mmol), CuI (0.005 mmol), and Cs_2CO_3 (0.4 mmol) in THF (2.0 mL) at 50 °C under an Ar atmosphere for 4.0 h. All the yields were the isolated yields. n. d. = not detected.

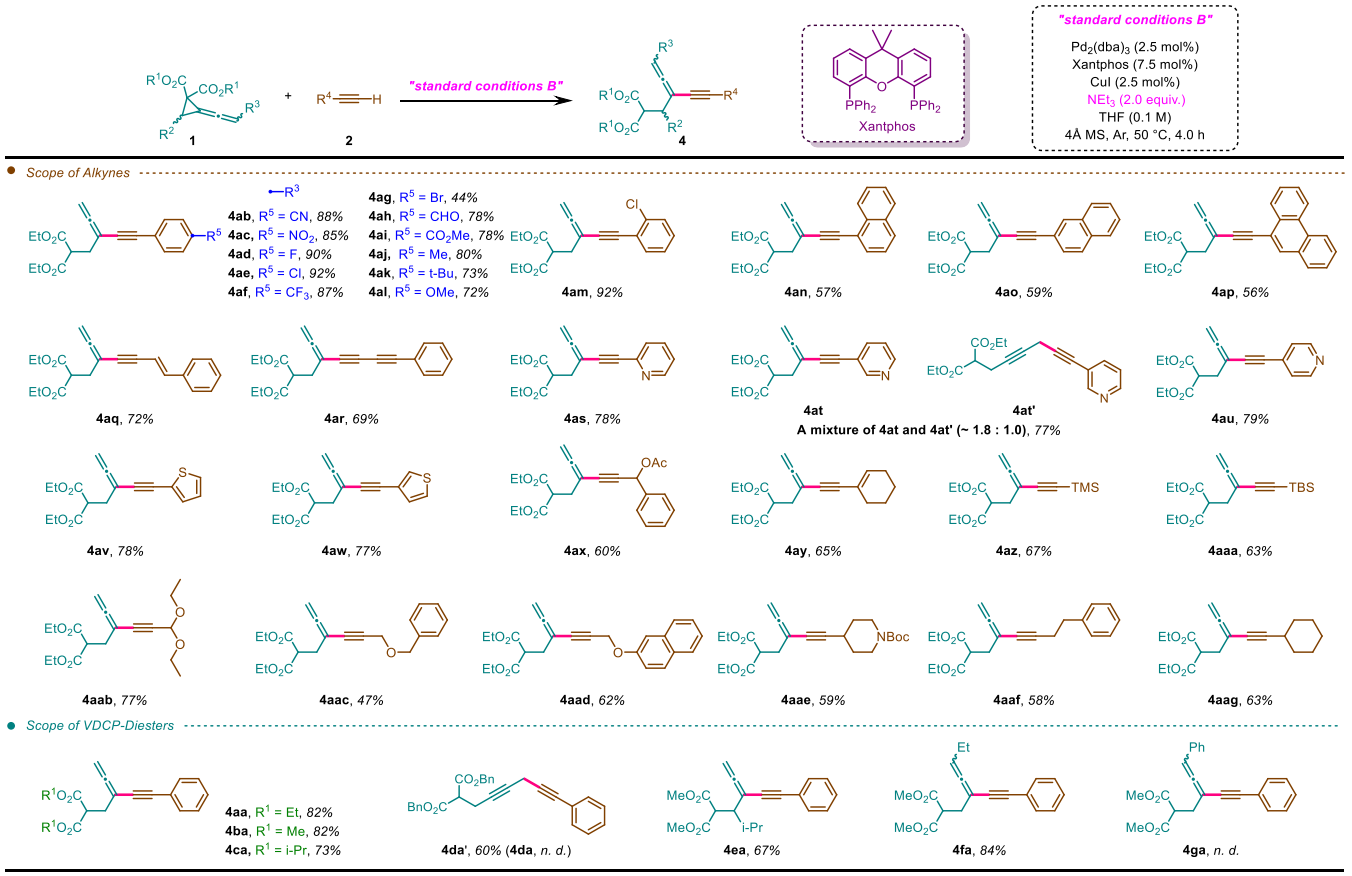
purified by flash silica gel chromatography (eluent: hexane/ethyl acetate = 20:1−6:1, v/v) to afford the desired products **3** or **4**.

RESULTS AND DISCUSSION

To initiate our studies, we started with a model reaction using vinylidenecyclopropane-diethyl ester **1a** (1.0 equiv) and phenylacetylene **2a** (1.5 equiv) as the coupling partners to optimize the reaction conditions. A bimetallic catalyst system consisting of the Pd(0) complex $Pd(PPh_3)_4$ and Cu(I) salt CuI was chosen. Meanwhile, an inorganic base Cs_2CO_3 (1.0 equiv) and 4 Å MS (100 mg) were used to perform this C−C cross-coupling reaction in the presence of $Sc(OTf)_3$ (10 mol %). Delightfully, this reaction proceeded smoothly in 2.0 mL of dry THF under the protection of argon at 50 °C for 4.0 h, giving the two different products **3aa** and **4aa** with low yields and poor chemoselectivity control (Table 1, entry 1). Next, we examined the equivalent of base effect and found that 2.0 equiv of base is the most appropriate, and the Lewis acid has a certain inhibitory effect on this reaction through control experiments (Table 1, entries 2−5). When no base was used, the reaction would not occur at all, and the equivalent of base of more than 2.0 was not economical (see Table S1 in the Supporting Information). Subsequently, we optimized the selection of ligands and used $Pd_2(dba)_3$ as the palladium source (see Table S2 in the Supporting Information). Upon examining several different phosphine ligands on the reaction outcome, we found that the aryl-substituted bisphosphine ligands, such as dppf, BINAP, and DPEphos, can obtain two products in medium yields with lower selectivity. When the bidentate phosphine ligand Xantphos was used, the product **3aa** was obtained exclusively with high selectivity (Table 1, entry 8, 85% NMR yield, 83% isolated yield). Two alkyl-substituted bisphosphine ligands dppp and dppb can also gain the single cyclic 1,3-enyne product **3aa** in NMR yields of 60% and 63%, respectively. Besides, the use of racemic phosphine-oxazoline bisphosphine ligand Ph-PhOX can also deliver a single product **3aa** in 44% NMR yield. However, when using monophosphine ligand with large steric hindrance, the NMR yields of both two products were not high. The other palladium source has also been examined to study the valence state of metal palladium, and the result showed that $Pd(OAc)_2$ and PPh_3 instead of $Pd(PPh_3)_4$ can afford **3aa** and **4aa** in 52% and 12% NMR yields, respectively (Table 1, entry 6), while using $Pd(OAc)_2$ itself as the catalyst, no reaction occurred (Table 1, entry 7). This observation may be due to the fact that an excess of PPh_3 ligand can reduce the bivalent palladium to zerovalent in the reaction system, suggesting that Pd(0) is the key catalyst to initiating the reaction. The types of different bases were likewise studied. On the basis of a throughout evaluation of the aforementioned factors, we have preliminarily identified the optimal conditions under standard condition A

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Scheme 3. Substrate Scope of Cross-Coupling Reactions of Vinylidenecyclopropane-Diesters 1 with Terminal Alkynes 2 under the Standard Conditions B



[a]All reactions were carried out with **1** (0.2 mmol), **2** (0.3 mmol), $Pd_2(dba)_3$ (0.005 mmol), Xantphos (0.015 mmol), CuI (0.005 mmol), and dry NEt_3 (0.4 mmol) in THF (2.0 mL) at 50 °C under an Ar atmosphere for 4.0 h. All the yields were the isolated yields. n. d. = not detected.

for the highly specific generation of **3aa** (Table 1, entry 8). Gratifyingly, the corresponding allenyne product **4aa** was obtained as a single one in the presence of an organic base such as triethylamine (Table 1, entry 9, 90% NMR yield, 82% isolated yield). After screening a large number of organic and inorganic bases (see Table S3 in the Supporting Information), we found that the use of cesium carbonate and triethylamine, respectively, provides good control on the chemoselectivities and exclusively generates cyclic 1,3-enyne **3aa** and allenyne **4aa**. Copper salts with different counterions were also screened, and among these copper salts used, CuI was identified as the most suitable one in this divergent synthesis of **3aa** and **4aa** (see Table S4 in the Supporting Information). The examination of the valence state of copper salts revealed that monovalent copper was the suitable choice. It is notable that the reaction can still proceed successfully when catalytic quantities of copper phenylacetylide are employed (Table 1, entries 10−11). In addition, the mechanistic function of Cu(I) was also investigated. In the presence of Cs_2CO_3, using $Pd_2(dba)_3$ as the palladium source and Xantphos as a ligand under otherwise identical conditions, we replaced the copper source with the different metal sources, such as $AgNTf_2$ and PPh_3AuNTf_2 as the cocatalyst to perform the reaction, and found that although they also have good selective control of the reaction outcome, the NMR yields of **3aa** decreased to only 34% and 44%, respectively (Table 1, entries 12−13). It is worth noting that we specifically investigated the reaction without the addition of copper catalyst CuI (Table 1, entry 14), palladium catalyst $Pd_2(dba)_3$ (Table 1, entry 15), and ligand Xantphos (Table 1, entry 16) to verify the crucial role of each component in the catalytic system. The results indicated that regardless of the absence of any one of the above three, the reaction was hardly able to proceed smoothly. This fully proved the irreplaceability of Pd, Cu, and ligand in this reaction, and the synergy effect among the three was a necessary condition for the smooth occurrence of the reaction.

After careful investigation of other conditions such as reaction temperature and solvent (see Tables S5−S8 in the Supporting Information), we finally determined the optimal conditions to access the desired products, which were identified as those described in entry 8 for **3aa** (standard conditions A) and entry 9 for **4aa** (standard conditions B) (for more information, see Tables S1−S8 in the Supporting Information). Furthermore, we have further optimized the reaction conditions relying solely on copper catalysis to explore the possibility of directly obtaining the allenyne product **4aa** or the 1,4-diyne product **SI-11** shown in Table S9 in the Supporting Information (based on Chen's work[61−63]) without introducing other catalysts. Through systematic screening of key parameters, such as base, temperature, reaction time, and solvent, the results demonstrated that the reaction failed to proceed as anticipated. This could be attributed to the remarkable reversibility of the copper acetylide species formed from the unactivated terminal alkyne during the reaction

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

process. Simultaneously, the capture process between this intermediate and the VDCP-diesters possesses a relatively high energy barrier, thereby causing the reaction to occur with difficulty (for a more detailed discussion, please refer to the Supporting Information on pages S13−S14).

On the basis of the above optimized conditions for this Pd/Cu-cocatalyzed divergent synthetic protocol, the substrate scope was investigated, and the results are displayed in Schemes 2 and 3.

To begin with these investigations under standard conditions A, we studied the application scope of unactivated terminal alkyne substrates in the reaction with VDCP-diester **1a**. For phenylacetylene derivatives with substituents on the benzene ring, cyclic 1,3-enyne products **3ab**−**3am** can be obtained in moderate to excellent yields. However, groups with electron-withdrawing effects often had higher yields than those of substrates with electron donors (**3ab**−**3af**, **3ah**, and **3ai** compared with **3aj**−**3al**). This may be attributed to the fact that cyclization is an intramolecular nucleophilic addition process. The substituents at the para-position of the benzene ring influence the electron cloud density of the allenic carbon−carbon double bond through the extensive conjugated system, and the reactivity of relatively electron-deficient allenes is higher. A special case of *para*-bromine-substituted phenylacetylene (**2g**), which had a yield of only 46%, was probably due to the side effects of other C−C cross-couplings under the reaction conditions. In addition, the electronic effects of *ortho*-chlorine-substituted and *para*-chlorine-substituted phenylacetylene are similar, and their yields (97%, 96%) were not much different. In addition to the benzene ring, other aromatic rings such as naphthalene and phenanthrene-derived terminal yne substrates are also well adapted to this transformation reaction, and the corresponding target products **3an**−**3ap** can be obtained in yields of 58%−61%. If the conjugated system of the substrate is extended with a carbon−carbon double bond (**2q**) or triple bond (**2r**) between the aromatic ring and the terminal alkyne, the reactions can still occur relatively well, affording products **3aq** and **3ar** in yields of 73% and 71%, respectively. In addition, the aromatic heterocycle-derived terminal alkyne substrates, including pyridine and thiophene (**2s**−**2w**), are also compatible with this Pd/Cu bimetallic synergistic catalytic system, the relative position of the heteroatoms on the ring has little effect on the yield of the reaction, and the target products **3as**−**3aw** can be obtained in the yields of 79%−86%. One interesting case has also been tested, in which substrate **2x** is commonly used as a copper-catalyzed propargylic substitution reaction. Despite this competing reaction, we found that the cyclic 1,3-enyne product **3ax** was still obtained as the main product under standard conditions A in 67% isolated yield. Several alkyl-substituted terminal alkyne substrates have also been tried, including alkyne substrate (**2y**), TMS (**2z**) or TBS-protected (**2aa**) alkynes, acetal (**2ab**), alkyl ether (**2ac**), alkyl aryl ether (**2ad**), Boc-protected piperidine ring (**2ae**), ethyl (**2af**), and cyclohexyl (**2ag**), all of which have been shown to be suitable for this conversion reaction. The yields of the desired products ranged from 42% to 76% (**3ay**−**3aag**), reflecting the wide applicability of the reaction substrates.

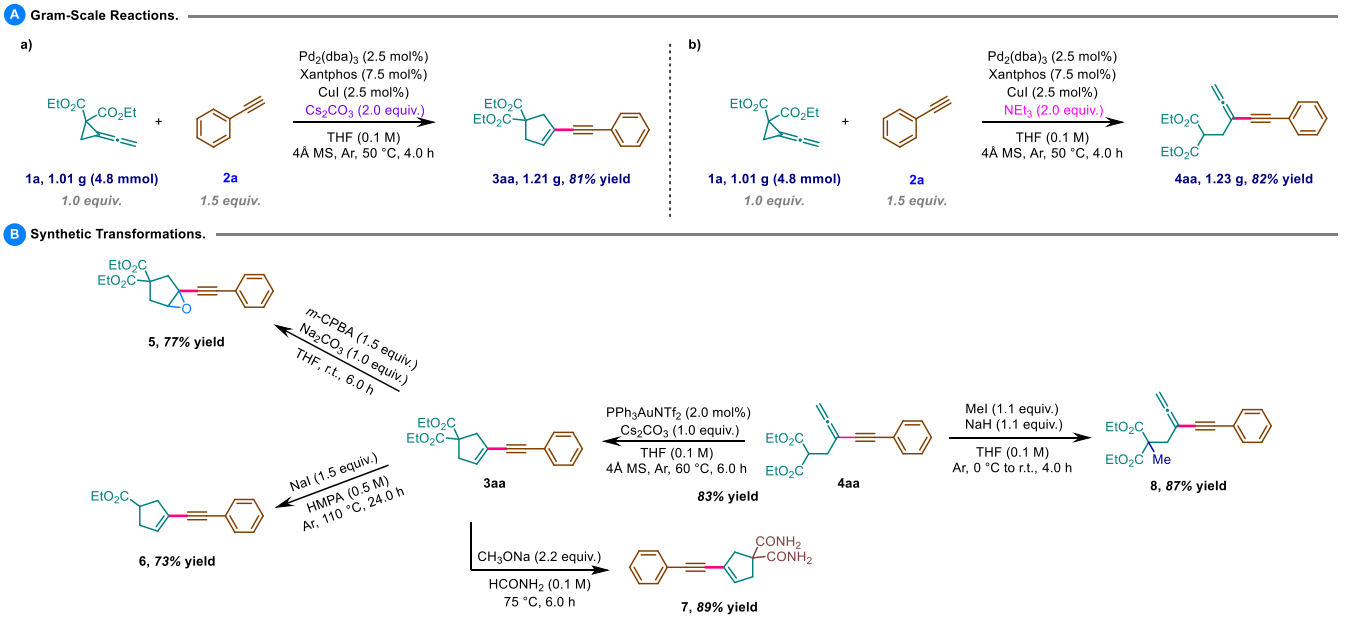
Subsequently, we set out to investigate a wide range of VDCP-diesters upon reacting with **2a** under the standard conditions A. Compared to VDCP-diester **1a**, reaction of **1b** with a dimethyl ester group with **2a** also proceeded smoothly, affording the corresponding product **3ba** in 84% yield. Increasing the steric hindrance of ester groups, such as VDCP-diester **1c** bearing a diisopropyl ester group and VDCP-diester **1d** with dibenzyl esters, the corresponding products **3ca** and **3da** were obtained in 70% and 59% yields, respectively, suggesting that the steric effect of VDCP-diesters indeed has some impact on the reaction outcome. If there is an isopropyl substituent on the three-membered ring of VDCP (**1e**), the reaction was specific to obtain an allenyne-type product **4ea**, probably due to the substituent on the side not being conducive to the intramolecular ring closing process. For the VDCP-diester with ethyl substitution at the terminal of the allene moiety, the reaction can also proceed smoothly to obtain the corresponding cyclic 1,3-enyne product **3fa** in 85% yield; when the ethyl group here is replaced with a phenyl group, the target product **3ga** cannot be obtained. Finally, we obtained product **3db** in 72% yield using substrates **1d** and **2b**, and the structure of the product was also confirmed by an X-ray single-crystal diffractometer.

Next, the scope of the cross-coupling reaction under the optimal standard conditions B for allenyne-type products was also investigated. Similarly, we first looked at the scope of application of terminal alkyne substrates on the basis of reactions with **1a** and found that varying different substituents at the para- and ortho-position on the benzene ring afforded the corresponding products **4ab**−**4am** in 44−92% yields. However, the yield was also lower when the para-position was bromine substituted, giving **4ag** in only a 44% yield. Moreover, larger conjugated systems, including naphthalene (**2an** & **2ao**), phenanthrene (**2ap**), and inserted carbon−carbon double bond (**2aq**) or carbon−carbon triple bond (**2ar**), were also tolerated in this reaction, which can also harvest the target products **4an**−**4ar** in 57%−72% yields. The pyridine ring and thiophene ring substituted at different positions can also be well adapted to the reaction conditions, and the corresponding target products **4as**−**4aw** (except **4at**) can be obtained in yields of 60% to 79%, respectively. It is important to note that when 3-ethynylpyridine **2t** was used as the coupling component, the reaction resulted in a mixture of allenyne **4at** and 1,4-diyne **4at′** (ca. 1.8:1.0, 77% total yield), and it is speculated that **4at′** was formed because of the presence of the isomerization of zwitterionic η^1-(allenyl)palladium species into η^3-(propargyl)palladium species in the reaction (see Scheme S21 on page S29 in the Supporting Information). In addition, the terminal alkynes substituted by the alkyl and silyl protective groups can also be used as good coupling substrates for this reaction, and the corresponding target products **4ax**−**4aag** can be obtained in 47% to 77% yields.

In addition, changing the ester groups in VDCP-diesters **1** from a dimethyl ester group to a diisopropyl ester group to react with **2a** also smoothly delivered the corresponding allenyne products **4ba**−**4ca** in 60% to 82% yields. In particular, when using the dibenzyl ester group-substituted VDCP substrate **1d** as the substrate, the reaction did not give an allenyne product **4da**, but a 1,4-diyne-type product **4da′** was produced, which may be caused by the isomerization of zwitterionic η^1-(allenyl)palladium species into η^3-(propargyl) palladium species (see Scheme S22 on page S30 in the Supporting Information). The addition of isopropyl substituent to the three-membered ring or ethyl substituent to the terminal of allene moiety also reacted well with phenylacetylene **2a**, giving the desired product **4ea** or **4fa** in 67% or 84% yield. However, the product **4ga** with a phenyl substitution at the terminal position of allenyne cannot be

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Scheme 4. Scale-Up Reactions and Synthetic Transformations. (A) Gram-Scale Reaction. (B) Synthetic Transformations



obtained, and it may be due to the fact that the steric hindrance is too large and is not conducive to the formation of the key intermediate.

To illustrate the synthetic utility of this Pd/Cu dual-catalyzed cross-coupling synthesis of vinylidenecyclopropane-diesters, gram-scale reactions have been conducted. We used VDCP-diester **1a** (1.01 g, 4.8 mmol) and phenylacetylene **2a** (0.74 g, 7.2 mmol) under the two different standard conditions A and B. The results show that the scale-up syntheses were performed successfully to verify the utility of the reaction, affording the corresponding products **3aa** or **4aa** in 81% and 82% yields, respectively (Scheme 4A).

Moreover, some further transformations of the obtained products **3aa** and **4aa** have also been investigated (Scheme 4B). On the one hand, cyclic 1,3-enyne **3aa** can achieve highly selective epoxidation of the olefinic fraction upon treating with 3-chloroperoxybenzoic acid (*m*-CPBA) and furnish the epoxy compound **5** in 77% yield. Besides, one of the two ester groups in substrate **3aa** can be efficiently removed at a high temperature to gain the single product **6** with 73% yield. In addition, the diethyl ester group can be converted into other substituents such as primary amide **7**, which is generally difficult to achieve in the synthesis of the corresponding VDCP substrate. On the other hand, for the allenyne **4aa**, it can undergo an intramolecular nucleophilic addition reaction to close the ring under the catalysis of Au(I) metal salt, with a yield of 83% to generate the product **3aa** under mild reaction conditions. An α-alkylation reaction can be carried out under the action of strong base NaH, and the methylated product **8** can be obtained in 87% yield when MeI was used as an electrophile.

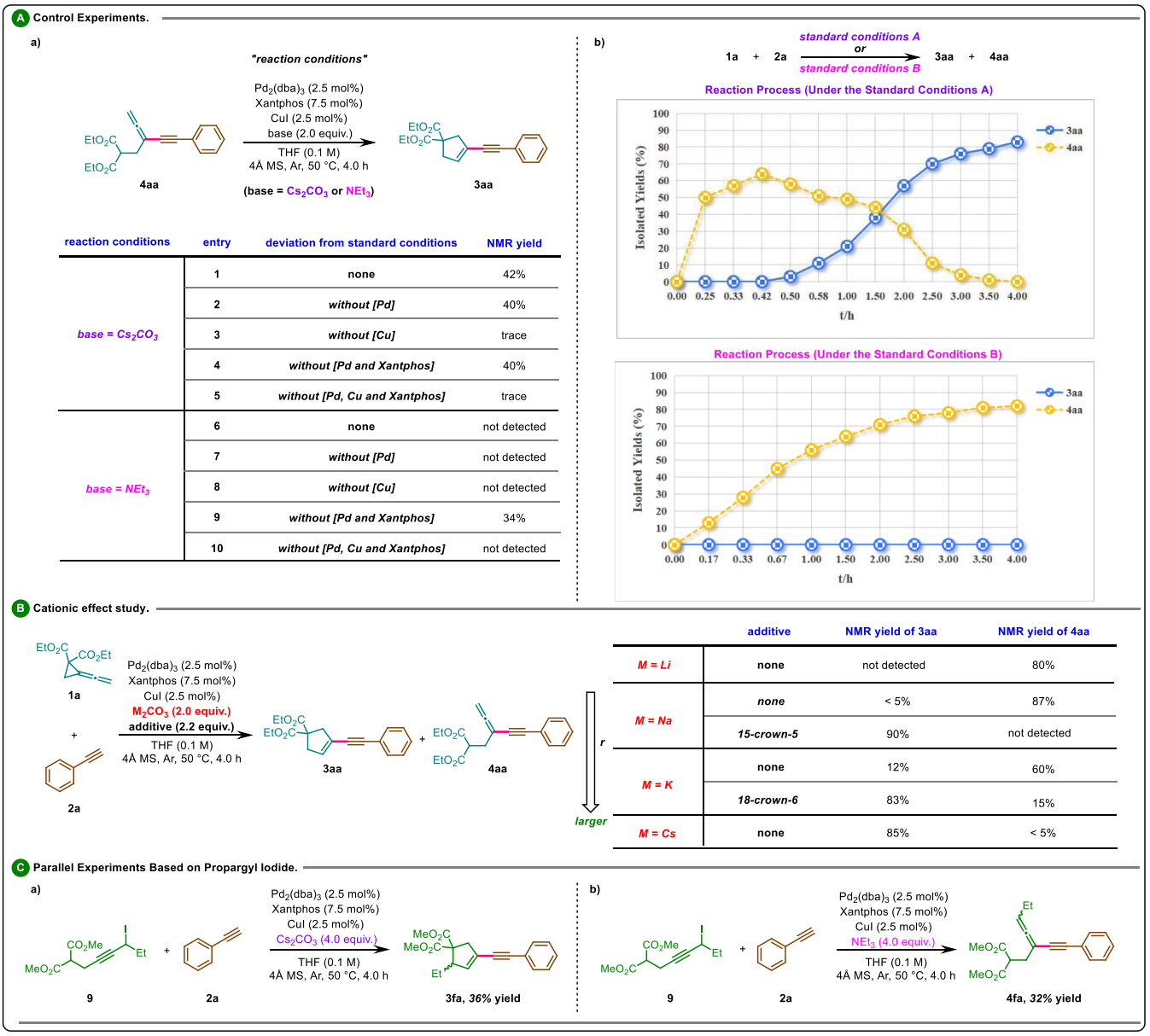
In order to validate the reaction mechanism, we performed a series of mechanistic experiments. We initially attempted to explore the potential active intermediates present in the reaction system. An excess of copper phenylacetylene **SI-12** was employed in lieu of the catalytic amount of cuprous iodide, and phenylacetylene was not added simultaneously. The reactions were conducted under standard conditions A or B, respectively (Scheme S8a,b in the Supporting Information); however, they did not occur successfully. It was hypothesized that this feeding manner inhibited the hydrogen transfer process in the reaction. To further validate our speculation, $CsHCO_3$ was used to replace Cs_2CO_3 under standard conditions A, and $NEt_3{\cdot}HCl$ was used to replace NEt_3 under standard conditions B. Through the additional introduction of a proton source, it was confirmed that the reaction could proceed smoothly, enabling the specific acquisition of the corresponding products with yields of 18% and 11%, respectively (Scheme S8c,d in the Supporting Information). This result suggests that the proton transfer process during the reaction is essential. The low yield might be attributed to excessive proton source leading to the occurrence of other side reactions and reducing the alkalinity of the reaction system.

To further elucidate the proton transfer pathway in the reaction, deuterium incorporation experiments were subsequently conducted under standard conditions A and B, as illustrated in Scheme S10 in the Supporting Information. The use of homemade deuterated **2a**-*d* as a substrate to react with VDCP-diester **1a** afforded **3aa**-*d* in 83% yield along with 29% deuteration and **4aa**-*d* in 80% yield along with 26% deuteration, respectively. The low deuterium substitution ratio in the products can be attributed to the rapid hydrogen−deuterium exchange that occurs between the deuterated protons and the ambient water or moisture that is difficult to completely remove from the reaction system. Based on the previous studies, we speculate that there is a coordination of Cu and a carbon−carbon triple bond in phenylacetylene in the reaction, and the hydrogen atom at the terminal position of phenylacetylene is deprotonated by the base in the system after activation and then enters into the product molecule through the subsequent transformation process.

Furthermore, several control experiments based on the conversion of allenyne product **4aa** to cyclic 1,3-enyne product **3aa** were also conducted (Scheme 5A(a)). We initially investigated the relevant reaction variables under standard

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Scheme 5. Mechanistic Studies. (A) Control Experiments. (B) Cationic Effect Study. (C) Parallel Experiments Based on Propargyl Iodide



reaction conditions	entry	deviation from standard conditions	NMR yield
base = Cs_2CO_3	1	none	42%
	2	*without [Pd]*	40%
	3	*without [Cu]*	trace
	4	*without [Pd and Xantphos]*	40%
	5	*without [Pd, Cu and Xantphos]*	trace
base = NEt_3	6	none	not detected
	7	*without [Pd]*	not detected
	8	*without [Cu]*	not detected
	9	*without [Pd and Xantphos]*	34%
	10	*without [Pd, Cu and Xantphos]*	not detected

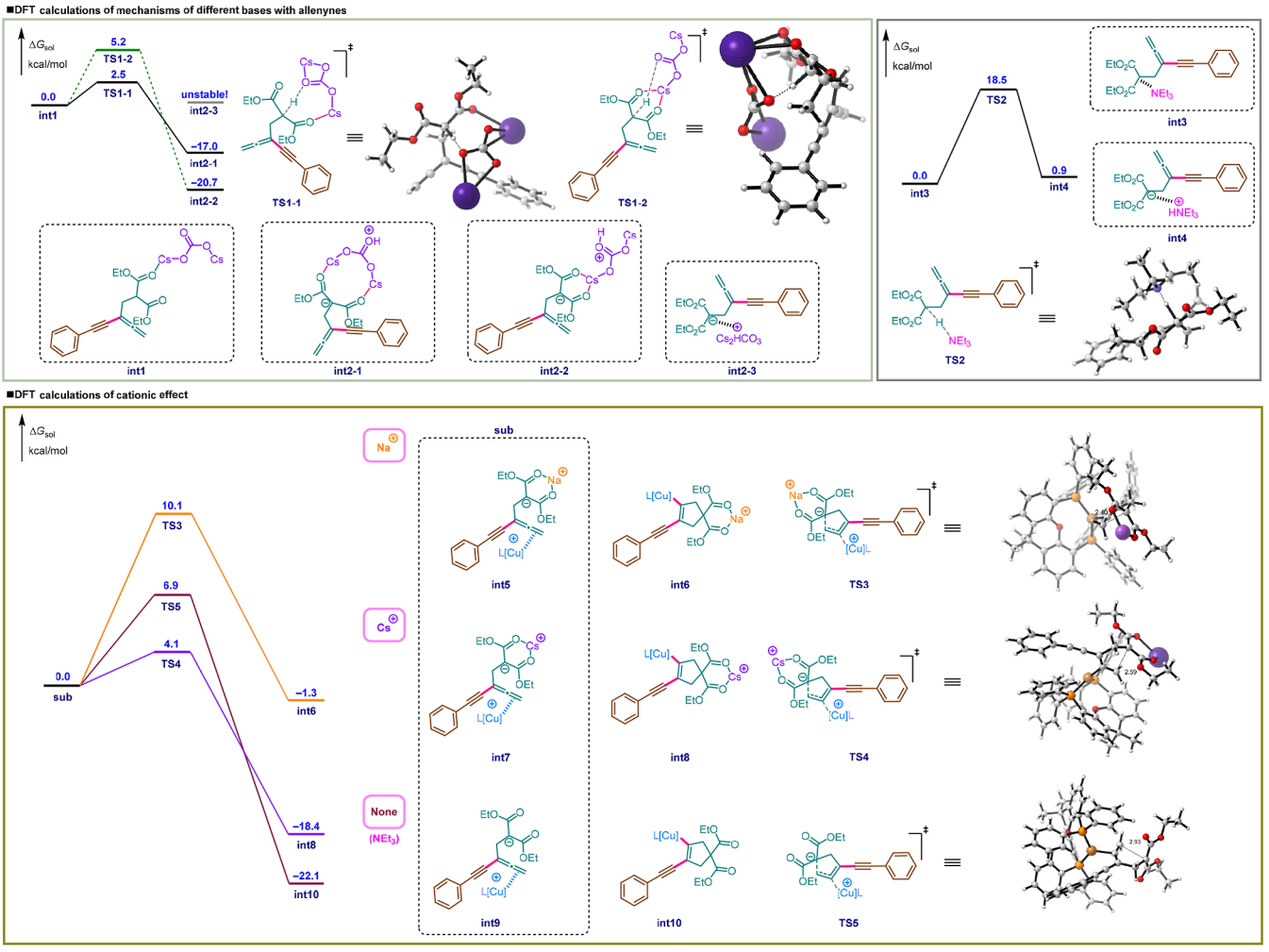
	additive	NMR yield of 3aa	NMR yield of 4aa
M = Li	none	not detected	80%
M = Na	*none*	< 5%	87%
	15-crown-5	90%	not detected
M = K	none	12%	60%
	18-crown-6	83%	15%
M = Cs	none	85%	< 5%

conditions A. The experimental results demonstrate that under unchanged conditions, the cyclization reaction proceeds smoothly with a yield of 42% for **3aa** (entry 1). Notably, the reaction can still proceed normally in the absence of palladium (entry 2). However, when copper was omitted, the reaction is difficult to occur (entry 3). Interestingly, in the absence of both palladium and the Xantphos ligand, the reaction also proceeds smoothly, leading to the formation of **3aa** in 40% yield (entry 4). In contrast, the reaction almost cannot take place when only cesium carbonate was added (entry 5). Based on the above experimental results, we preliminarily speculate that the copper catalyst plays a crucial role in the cyclization process. On the other hand, in control experiments using triethylamine as the base, we observed that the reaction did not proceed under standard condition B (entry 6) or in the absence of palladium source (entry 7) and copper source (entry 8). However, when both the palladium source and Xantphos were omitted simultaneously, the reaction unexpectedly proceeded, affording **3aa** in a yield of only 34% (entry 9). This result further confirms that Xantphos exerts an inhibitory effect on the cyclization reaction, and this inhibition is more pronounced in the presence of NEt_3 rather than Cs_2CO_3 as the base. Similarly, the reaction did not occur when only triethylamine was added without any additional catalysts (entry 10), and this observation further clarifies that the cyclization reaction indeed requires the participation of a copper catalyst and Cs_2CO_3. In a short summary, the interaction between different bases and copper coordinated with phosphine ligands influences the reaction process.[64]

Next, we carried out a more comprehensive analysis of the corresponding reaction processes under the two reaction conditions through the designed controlled experiments (Scheme 5A(b)). Under standard conditions A, it was observed that the reaction initially proceeded rapidly to form the allenyne product **4aa**. However, upon reaching a certain conversion rate (t = 25 min), the concentration of **4aa** begins

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ACS Catal. 2025, 15, 8390−8402

Scheme 6. DFT-Computed Energy Profile for Synergistic Pd/Cu-Catalyzed Cross-Coupling Reactions of Vinylidenecyclopropane-Diester 1a and Phenylacetylene 2a. Free Energies Are Calculated at the SMD(tetrahydrofuran)/B3LYP(D3BJ)/def2-tzvp//B3LYP(D3BJ)/6-31G(d,p)/SDD Level Using the Gaussian 16 Program



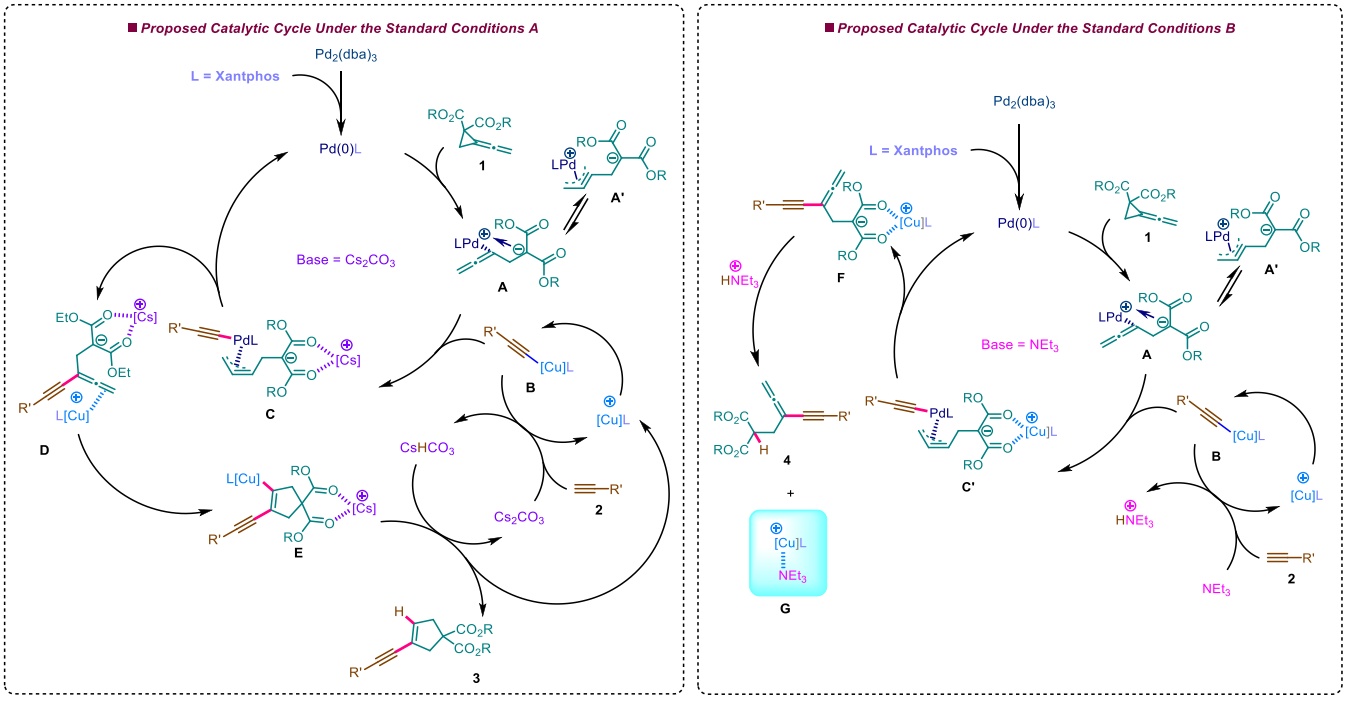
to decrease as the reaction shifts toward the production of 1,3-enyne product **3aa**, ultimately leading to the complete transformation of **4aa** into **3aa**. Additionally, it was noted that the formation rates of both **4aa** and **3aa** exhibit an initial rapid phase followed by a gradual slowdown, which might be attributed to the fact that during the initial stage, the concentration of the relevant raw materials was maintained at a relatively high level. Similarly, under standard conditions B, the reaction exclusively produces **4aa** throughout, with its formation rate also following a pattern of being initially fast and subsequently slowing down.

In an attempt to further validate the impact of cationic effects on the chemical selectivity of the reaction, we carried out a systematic investigation into the impact of various alkali metal carbonates on the reaction and the Lewis acidity of alkali metal ions, with the results presented as shown in Scheme 5B. The results show that when using lithium salt (Li_2CO_3), a selective generation of allenyne-type product **4aa** occurred, and as the ionic radius increases such as using sodium carbonate (Na_2CO_3) as a base, a very small amount of cyclic 1,3-enyne-type product **3aa** was generated. At this point, if 15-crown-5 was added to the complex with sodium ion, the reaction selectivity will be completely reversed, giving **3aa** in 90% yield. Using potassium carbonate (K_2CO_3) as a base, which has an even bigger ionic radius, the trend of the reaction toward the generation of **3aa** was slightly greater than those of using Li_2CO_3 or Na_2CO_3 as a base, and when using cesium carbonate as a base, almost none of allenyne product **4aa** was detected. Similarly, when potassium ions form a complex with 18-crown-6, the reaction exhibited a significantly increased propensity to produce the cyclic 1,3-enyne product **3aa** in 83% yield, which is considerably higher than the 12% observed in the control experiment. Compared to the scenario where cesium carbonate serves as a proton transfer shuttle, we hypothesize that the cesium ionic effect has a significant influence on the product selectivity. The density functional theory (DFT) calculations of the cationic effect will further clarify the reaction mechanism (for details, see Scheme 6 and the related discussion).

Ultimately, based on the results in the conditional screening, the palladium metal species are also important for the occurrence of this reaction. In our previous study, we generally believed that the palladium catalyst first undergoes an oxidative addition process with the VDCP-diesters molecule to obtain the zwitterionic η^1-(allenyl)palladium species, and in this protocol, we used a propargyl iodine **9** instead of VDCP-diester **1f** to investigate the feasibility of the reactions (Scheme 5C(a,b)). To neutralize the acid in situ generated in the

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

Scheme 7. Proposed Mechanisms for the Productions of Cyclic 1,3-Enynes 3 and Allenynes 4



reaction system, we treated it with an excess of base such as using 4.0 equiv of Cs_2CO_3 and NEt_3. The palladium catalyst is generally thought to be able to undergo an oxidative addition with this propargyl iodine substrate and then in situ generate the η^3-(propargyl) palladium species in the presence of a base in the system, which can confirm the isomerization of the zwitterionic η^1-(allenyl)palladium species and the zwitterionic η^3-(propargyl)palladium species present in this reaction. As a matter of fact, the corresponding products **3fa** and **4fa** were produced in 36% and 32% yields, respectively, rendering the presence of a zwitterionic palladium species in the reaction system.

Furthermore, DFT calculations of some key steps were performed to gain more insight into the reaction mechanism (Scheme 6). The primary distinction lies in the differences between the two distinct bases involved in the deprotonation process of the key intermediate. In order to explore the interaction mechanisms of different bases with the allenyne, we conducted DFT calculations. In the case of cesium carbonate, due to the certain Lewis acidity of the cesium ion, several distinct interaction patterns occur between the cesium ion and the malonate moiety of the allenyne, including the malonate serving as a monodentate or bidentate ligand of the cesium ion. Nevertheless, for the process involving triethylamine, the trimethylamine only plays a role as a Brønsted base. The results indicate that in THF solvent, cesium carbonate effectively facilitates the α-deprotonation of the malonate in allenyne **4aa** to form an intermediate **int2-1** or **int2-2** as a strongly exothermic process (**int2−1**: $\Delta G = -17.0$ kcal·mol^{-1}, **int2-2**: $\Delta G = -20.7$ kcal·mol^{-1}). The energy barrier for deprotonation is 2.5 kcal·mol^{-1} through the transition state **TS1-1**, which is lower than that of another pathway passing through the transition state **TS1-2** ($\Delta G^{\ddagger} = 5.2$ kcal·mol^{-1}) due to different coordination modes. The deprotonation of **4aa** utilizing trimethylamine as a base is a slightly endothermic process ($\Delta G = 0.9$ kcal·mol^{-1}), and the energy barrier of deprotonation is 18.5 kcal·mol^{-1} via a transition state **TS2**, which is much higher than that using cesium carbonate as a base. Therefore, the deprotonation of triethylamine is both thermodynamically and kinetically less favorable. This phenomenon can be attributed to the higher basicity of cesium carbonate compared to triethylamine in THF. The calculation results rationalize why the product **4aa** was obtained experimentally in the case of using triethylamine as a base (see the catalytic cycles under the standard conditions A and B in Scheme 7).

In addition, the influence of cations on the reaction outcomes, which has been identified during the reaction condition optimization (see Table 1 and the related results summarized in the Supporting Information), is investigated through DFT calculations. It is evident that the cationic effect induced by sodium ion coordination results in an energy barrier of 10.1 kcal·mol^{-1} passing through the ring-closing transition state **TS3**, accompanied by a slightly exothermic process ($\Delta G = -1.3$ kcal·mol^{-1}). Comparably, the cesium ion-induced cationic effect reduces this energy barrier by 6.0 kcal·mol^{-1} (**TS4**) and is associated with a significantly exothermic process ($\Delta G = -18.4$ kcal·mol^{-1}). Additionally, we have calculated the scenario without metal ion coordination, revealing that in the absence of any metal cation coordination, the energy barrier for the ring-closing process is 6.9 kcal·mol^{-1} (**TS5**). Although this value is lower than that observed with sodium ions, it remains higher than that of transition state **TS4** when cesium ions are present. By comparison of the energy barriers of the transition states during the ring closure process with those of the substrate lacking metal ion coordination, it is evident that the cationic effect induced by sodium ions elevates the corresponding transition state energy barrier, thereby favoring the retention of the reaction at the allenyne stage. It is hypothesized that the α-carbanion of malonate becomes stabilized upon coordination with sodium ions and the diester group, leading to a reduction in nucleophilicity, which hinders

https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

ring closure. Conversely, the energy barrier of the transition state decreases due to the cesium ionic effect exerted by cesium ions, facilitating the progression of the reaction toward the cyclic 1,3-enyne product. This cesium-induced effect on the reaction dynamics aligns with findings from prior investigations.[65−67]

On the basis of the previously reported processes, our experimental research mentioned above, and DFT calculations, the plausible mechanisms of this divergent synthesis protocol have been depicted in Scheme 7. The reaction begins with Pd(0), and the precatalyst $Pd_2(dba)_3$ undergoes a ligand exchange with Xantphos to obtain a reactive Pd(0)L catalyst (L = Xantphos), and then an oxidative addition of Pd(0)L to VDCP-diester **1** takes place to give the corresponding zwitterionic η^1-(allenyl)palladium species **A**, which can undergo isomerization to η^3-(propargyl)palladium species **A′**; at the same time, the copper catalyst activates terminal alkynyl hydrogen atom of **2** through coordination and forms a copper acetylide species **B** through the action of base (Cs_2CO_3), and the two species achieve the formation of key intermediate **C** through a rapid process of transmetalation. Subsequently, intermediate **C** undergoes a reductive elimination along with regeneration of the Pd(0) catalyst, affording intermediate **D**, which undergoes a Cu(I)-catalyzed cyclization, resulting in the cyclic 1,3-enyne product **3** through protonation via intermediate **E**, and the Cu(I) catalyst is regenerated. For the formation of the allenyne product **4**, analogous to the previously described process, a transmetalation process involving a zwitterionic η^1-(allenyl)palladium species **A** or η^3-(propargyl)palladium species **A′** with an in situ generated copper acetylide species **B** initially takes place, leading to the formation of species **C′**, followed by a reductive elimination process, which releases a Pd(0) catalyst and obtains the allenyne-type intermediate **F**. Then, **F** is rapidly quenched by protonated triethylamine (supported by DFT calculations of mechanisms of different bases with allenynes, see Scheme 6), thereby yielding the allenyne product **4**. In the presence of excess triethylamine, copper-amine complex **G** is also formed. In a short summary, since excess triethylamine plays a role as a stronger Lewis base and possesses a relatively weak deprotonation ability, it preferentially coordinates with copper(I) as complex **G**, resulting in a stagnation of the reaction at the allenyne stage. On the other hand, the nonhomogeneous proton transfer shuttle Cs_2CO_3 acts as a stronger Brønsted base, significantly enhancing the driving force for the cyclization reaction through its interaction with the substrate via the cesium ionic effect (supported by DFT calculations on the cationic effect, see Scheme 6), and the allenyne product **4** can be more easily further activated by copper(I) species, facilitating an intramolecular nucleophilic addition and subsequent ring closure to yield the cyclic 1,3-enyne product **3**.

CONCLUSION

In summary, a Pd/Cu dual catalysis and Cs_2CO_3 and NEt_3-mediated divergent synthesis of terminal alkynes with zwitterionic palladium intermediates generated from VDCP-diesters has been developed, obtaining two different structurally diverse products, cyclic 1,3-enynes and allenynes, with high research value. This divergent synthesis can gain various aromatic or alkyl structure substitution products in moderate to good isolated yields with good functional group compatibility. On the basis of our experimental results as well as DFT calculations, palladium is vitally important to the formation of key intermediate zwitterionic palladium species, and copper concurrently performs the functions of transmetalation and catalytic cyclization in the reaction, which significantly ameliorated the issue of low cyclization yield under the condition of single copper catalysis to a considerable extent, thereby manifesting the superiority of synergistic catalysis in one-pot. The employed bases such as cesium carbonate and triethylamine, which function as proton transfer shuttles, lead to divergent synthetic outcomes in the reaction due to their distinctly different Brønsted basicity and Lewis basicity. Moreover, the cesium effect is of vital significance for the smooth occurrence of the induced cascade reaction. In the whole process, the two different bases promote the transmetalation process and regulates the intramolecular nucleophilic addition by interacting with copper. In addition, as the key intermediates of chiral transformation and synthesis, cyclic 1,3-enynes and allenynes themselves have been reported to undergo interesting reactions as multifunctional synthons. Further research based on the products of this divergent cross-coupling reaction for the synthesis of chiral biologically active molecules are undergoing in our laboratory now. Moreover, considering that the chirality of the compound has attracted considerable attention, an asymmetric version of these reactions is also underway.

MATERIALS AND METHODS

All information pertaining to the materials and methods used in this study is reported in the Supporting Information.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c01895.

General procedures, characterization data, NMR spectra, and crystallographic data (PDF)

Computational details (CIF)

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https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We are grateful for the financial support from the National Key R & D Program of China (2023YFA1506700), the Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDB0590000), and the National Natural Science Foundation of China (21372250, 21121062, 21302203, 21772037, 21772226, 21861132014, 91956115, and 22171078).

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https://doi.org/10.1021/acscatal.5c01895
ACS Catal. 2025, 15, 8390−8402